THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 2: **Linear Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $p(x) = 3x - 1$, then $p(p(2)) =$ 1

(a) 13 **(b)** 15 **(c)** 16 **(d)** 14

2. A plan costs ₹50 plus ₹15 per unit; another costs ₹170 plus ₹5 per unit. They cost the same for ___ units. 1

(a) 11 **(b)** 13 **(c)** 14 **(d)** 12

3. If $x = 7$ is the zero of $kx - 56$, then $k =$ 1

(a) 8 **(b)** 7 **(c)** 10 **(d)** 9

4. The graph of $y = 7$ (a constant polynomial) is: 1

(a) a horizontal line **(b)** a vertical line

(c) a point **(d)** a line through the origin

5. For the pattern with nth term $5n + 2$, the sum of the 1st and 12th terms is: 1

(a) 70 **(b)** 69 **(c)** 71 **(d)** 68

6. For which value of n is $12n + 7 = 271$? 1

(a) 23 **(b)** 21 **(c)** 22 **(d)** 24

7. The area (in square units) of the triangle formed by the line $y = 6x + 24$ and the two axes is: 1

(a) 49 **(b)** 50 **(c)** 47 **(d)** 48

8. The x-coordinate of the point where $y = 6x + 4$ and $y = -x + 25$ meet is: 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

9. If $p(x) = 4x + 1$, then $p(p(3)) =$ 1

(a) 55 **(b)** 52 **(c)** 54 **(d)** 53

10. A plan costs ₹80 plus ₹25 per unit; another costs ₹200 plus ₹15 per unit. They cost the same for ___ units. 1

(a) 11 **(b)** 12 **(c)** 14 **(d)** 13

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. If $p(x) = 2x - 7$, find $p(p(x))$ and its value at $x = 5$. 2

12. Find the area of the triangle formed with the axes by the line $y = 4x + 20$. 2

13. Plan A costs ₹90 plus ₹40 per unit; Plan B costs ₹330 plus ₹20 per unit. For how many units do both cost the same? 2

14. If $p(x) = 4x + 3$, find $p(p(x))$ and its value at $x = 1$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the point of intersection of $y = 4x + 8$ and $y = -2x - 4$. 3

16. A linear polynomial has $p(1) = 18$ and $p(3) = 30$. Find $p(x)$, its zero and $p(3)$. 3

17. The nth term of a pattern is $16n + 5$. Find the number of terms that are less than 149 and the sum of the first two terms. 3

OR

Find the point of intersection of $y = 6x + 1$ and $y = x + 21$. 3

18. A linear polynomial has $p(1) = -18$ and $p(3) = -12$. Find $p(x)$, its zero and $p(10)$. 3

OR

The nth term of a pattern is $12n - 2$. Find the number of terms that are less than 178 and the sum of the first two terms. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two taxi services: A charges ₹50 plus ₹15 per km; B charges ₹110 plus ₹10 per km. Find the distance for which both charge equally, the common fare, and which is cheaper for 20 km. 5

OR

A line passes through $(0, 16)$ and $(3, 28)$. Find its slope, equation, intercepts and the area of the triangle it forms with the axes. 5

20. Two taxi services: A charges ₹20 plus ₹9 per km; B charges ₹80 plus ₹4 per km. Find the distance for which both charge equally, the common fare, and which is cheaper for 6 km. 5

OR

A line passes through $(1, -6)$ and $(5, 18)$. Find its slope, equation, intercepts and the area of the triangle it forms with the axes. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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